

Devops Day5

Devops class guvi (DAY-5)

21 March 2025

Granting Jenkins Sudo Privileges – The jenkins ALL=(ALL) NOPASSWD: ALL entry in the sudoers file allows the Jenkins user to run any command without a password prompt.

Restarting SSH Services – Commands like `sudo systemctl restart ssh.service` and `sudo systemctl restart sshd.service` restart the SSH service, ensuring remote login functionality.

Installing OpenSSH Server – The commands `sudo apt update` and `sudo apt install openssh-server` update package lists and install the OpenSSH server for secure remote access.

Checking SSH Service Status – `sudo systemctl status ssh` checks if the SSH service is running and displays its current status.

Systemd Service File Lookup – `ls /etc/systemd/system/sshd.service` or `ls /usr/lib/systemd/system/sshd.service` helps locate the SSH daemon's systemd service file.

Reloading Systemd Daemon – `sudo systemctl daemon-reload` ensures that systemd picks up changes in service configurations without requiring a reboot.

Encoding Minikube Certificate – `cat /home/david/.minikube/ca.crt | base64 -w 0; echo` encodes the Minikube CA certificate in base64 format, likely for authentication.

Changing Docker Socket Permissions – `sudo chmod 666 /var/run/docker.sock` grants read and write access to all users for Docker's Unix socket, allowing non-root users to interact with Docker.

Deploying Kubernetes Resources – `sh 'kubectl apply -f deployment.yml --validate=false'` applies a Kubernetes deployment file, ignoring validation errors.

Accessing Minikube Service – `minikube service my-service --url | xargs curl` retrieves the Minikube service URL and sends an HTTP request to test its accessibility.

Commands:

```
jenkins ALL=(ALL) NOPASSWD: ALL
```

```
sudo systemctl restart ssh.service
```

```
sudo systemctl restart sshd.service
```

```
sudo apt update
```

```
sudo apt install openssh-server
```

```
sudo systemctl restart ssh
```

```
sudo systemctl status ssh
```

```
ls /etc/systemd/system/sshd.service or ls /usr/lib/systemd/system/sshd.service
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl status ssh
```

```
sudo systemctl restart ssh.service
```

```
cat /home/david/.minikube/ca.crt | base64 -w 0; echo
```

```
sudo chmod 666 /var/run/docker.sock
```

```
[https://192.168.39.226:8443](https://192.168.39.226:8443/)
```

```
sh 'kubectl apply -f deployment.yml --validate=false'
```

```
minikube service my-service --url | xargs curl
```

Pipeline codes:

```
pipeline {
```

```
    agent any
```

```
    stages {
```

```
        stage('SCM') {
```

```
    steps {
        git branch: 'main', url: 'https://github.com/Pavish-kumar/guvidevopswarsday1'
    }
}

stage('Build-clean') {
    steps{
        sh 'mvn clean'
    }
}

stage('Build-validate') {
    steps{
        sh 'mvn validate'
    }
}

stage('Build-compile') {
    steps{
        sh 'mvn compile'
    }
}

stage('Build-test') {
    steps{
        sh 'mvn test'
    }
}

stage('Build-package') {
    steps{
        sh 'mvn package'
    }
}

stage('build to images') {
    steps {
    script{
```

```
    sh "docker build -t pavishkumar/webapplication ."
  }
}
stage('docker push hub') {
  steps {
    script{
      withDockerRegistry(credentialsId: 'cred-1', url: 'https://index.docker.io/v1/') {
        sh 'docker push pavishkumar/webapplication'
      }
    }
  }
}
stage('dep') {
  steps {
    script{
      withDockerRegistry(credentialsId: 'cred-1', url: 'https://index.docker.io/v1/') {
        sh 'docker push pavishkumar/webapplication'
      }
    }
  }
}
}
```

javaapplication [Jenkins] x Pipeline Syntax: Snippet x New cloud [Jenkins] x docker hub - Search x pavishkumar (Pavish Kum x +

localhost:8080/job/javaapplication/

Gmail YouTube GitHub Dashboard Cisco Networking A... TCS ION: Transform... Zoho Corp Careers

Jenkins

Dashboard > javaapplication >

Status ✓ javaapplication Add description

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Stage View

	SCM	Build-clean	Build-validate	Build-compile	Build-test	Build-package	build to images	docker push hub
Average stage times: (full run time: ~48s)	2s	3s	2s	3s	3s	4s	1s	6s
#3 13:40 No Changes	1s	942ms	1s	2s	2s	3s	1s	31s
#4 13:38 No Changes	1s	2s	2s	2s	2s	3s	1s	513ms failed
#5 13:37 No Changes	1s	1s	2s	2s	2s	4s	985ms	471ms failed

Builds

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```
pavis@pavish-kumar:~$ sudo systemctl restart ssh.service
pavis@pavish-kumar:~$ sudo systemctl daemon-reload
pavis@pavish-kumar:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2025-03-21 05:57:00 UTC; 1min 26s ago
 TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Main PID: 7269 (sshd)
     Tasks: 1 (limit: 4592)
    Memory: 1.2M (0)
   CGroup: /system.slice/ssh.service
           └─7269 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 21 05:57:00 pavish-kumar systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 21 05:57:00 pavish-kumar sshd[7269]: Server listening on :: port 22.
Mar 21 05:57:00 pavish-kumar systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
pavis@pavish-kumar:~$
```

Top Stories
Elon Musk's Pent...

```
ls: cannot access '/etc/systemd/system/sshd.service': No such file or directory
ls: cannot access 'or': No such file or directory
ls: cannot access 'ls': No such file or directory
ls: cannot access '/usr/lib/systemd/system/sshd.service': No such file or directory
pavis@pavish-kumar:~$ cd ~/.kube/
-bash: cd: ~/.kube/: No such file or directory
pavis@pavish-kumar:~$ cd ~
~_apt          ~games/       ~mail/        ~polkitd/     ~syslog       ~www-data
~backup/       ~irc          ~man/         ~proxy/       ~systemd-network/
~bin/          ~jenkins/    ~messagebus   ~root/        ~systemd-resolve/
~daemon/       ~landscape/  ~news        ~sshd/        ~systemd-timesync/
~dhcpcd/       ~list        ~nobody       ~sync/        ~uuwp
~dnsmasq/      ~lp          ~pavis/       ~sys/         ~uuidd/
pavis@pavish-kumar:~$ cd ~/.kube/
pavis@pavish-kumar:~/.kube$ ls
cache  config
pavis@pavish-kumar:~/.kube$ sudo nano config
pavis@pavish-kumar:~/.kube$ kubectl getnode
error: unknown command "getnode" for "kubectl"
pavis@pavish-kumar:~/.kube$ kubectl get node
NAME          STATUS    ROLES    AGE   VERSION
minikube     Ready    control-plane   2d    v1.32.0
pavis@pavish-kumar:~/.kube$
```

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```
pavis@pavish-kumar: ~/kube
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.8) ...
Selecting previously unselected package libwrap0:amd64.
Preparing to unpack .../2-libwrap0_7.6.q-33_amd64.deb ...
Unpacking libwrap0:amd64 (7.6.q-33) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../3-openssh-server_1%3a9.6p1-3ubuntu13.8_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.8) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../4-ncurses-term_6.4+20240113-1ubuntu2_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../5-ssh-import-id_5.11-0ubuntu2.24.04.1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.8) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libwrap0:amd64 (7.6.q-33) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.8) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.8) ...

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:rt+AZxNdi4psY1NFCG9uHfaZ5dh0Mdn03M+Yw08m3s root@pavish-kumar (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:xIjDfahxB1pg59wmgMnNRPj4KScgLD8rskoXtXVfH1A root@pavish-kumar (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:gEiYr6SwV/on0WVoTVmckY/FbVshUaU8d87Pb/RdCUE root@pavish-kumar (ED25519)
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
pavis@pavish-kumar:~$ sudo systemctl restart ssh.service
sudo systemctl restart sshd.service

Memory: 1.2M ()
CGroup: /system.slice/ssh.service
└─7269 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

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Mar 21 05:57:00 pavish-kumar systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
pavis@pavish-kumar:~$ ls /etc/systemd/system/sshd.service or ls /usr/lib/systemd/system/sshd.service
ls: cannot access '/etc/systemd/system/sshd.service': No such file or directory
ls: cannot access 'or': No such file or directory
ls: cannot access 'ls': No such file or directory
ls: cannot access '/usr/lib/systemd/system/sshd.service': No such file or directory
pavis@pavish-kumar:~$ cd ~/.kube/
-bash: cd: ~/.kube/: No such file or directory
pavis@pavish-kumar:~$ cd ~
~_apt          ~games/        ~mail/         ~polkitd/      ~syslog        ~www-data
~backup/       ~irc           ~man/          ~proxy/        ~systemd-network/
~bin/          ~jenkins/     ~messagebus    ~root/         ~systemd-resolve/
~daemon/       ~landscape/    ~news         ~sshd/         ~systemd-timesync/
~dhcpcd/       ~list         ~nobody        ~sync/         ~uuu           ~uuidd/
~dnsmasq/      ~lp           ~pavis/        ~sys/

pavis@pavish-kumar:~$ cd ~/.kube/
pavis@pavish-kumar:~/.kube$ ls
cache  config
pavis@pavish-kumar:~/.kube$ sudo nano config
pavis@pavish-kumar:~/.kube$ sudo nano config
pavis@pavish-kumar:~/.kube$ kubectl getnode
error: unknown command "getnode" for "kubectl"
pavis@pavish-kumar:~/.kube$ kubectl get node
NAME      STATUS    ROLES    AGE   VERSION
minikube  Ready    control-plane  2d   v1.32.0
pavis@pavish-kumar:~/.kube$
```


Dashboard > javaapplication > #5

```
bc05267c613b: Mounted from pavishkumar/guvideopswarsday1
5f70bf18a086: Mounted from pavishkumar/guvideopswarsday1
43c9f8a1dd61: Mounted from pavishkumar/guvideopswarsday1
b13564ac4148: Pushed
f844dcf94898: Mounted from pavishkumar/guvideopswarsday1
39cf0ac89a5a: Mounted from pavishkumar/guvideopswarsday1
3359bc3d7a6a: Mounted from pavishkumar/guvideopswarsday1
4b7c01ed0534: Mounted from pavishkumar/guvideopswarsday1
latest: digest: sha256:4443450117a96260baab998cdd97ca525a7c0e802cfd68b71d314899df323e4b size: 2409
[Pipeline] }
[Pipeline] // withDockerRegistry
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

REST API Jenkins 2.492.2

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javaapplication Config [Jen] x Pipeline Syntax: Snippet x New cloud [Jenkins] x docker hub - Search x Docker Hub

hub.docker.com/repositories/pavishkumar

pavishkumar Docker Personal

Repositories

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories

All repositories within the pavishkumar namespace.

Search by repository name All content Create a repository

Name	Last Pushed	Contains	Visibility	Scout
pavishkumar/webapplication	about 2 hours ago	IMAGE	Public	Inactive
pavishkumar/guvideopswarsday1	2 days ago	IMAGE	Public	Inactive
pavishkumar/warimage-jenkins	3 days ago	IMAGE	Public	Inactive
pavishkumar/hello-world	15 days ago		Public	Inactive

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